



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Project 3

Submission date: 16 May 2026

Group WO1

Student name	Student Number	Course Code
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Business Rules for the Application

Chat Participants

- Each user may join the live chat.
- Each user may leave the live chat.
- Each chat participant must have a user ID.
- Each chat participant must have a timestamp indicating when they joined.
- Each chat participant must have a chat ID.

Reviews

- Each user may leave one review per listing.
- A review may not be edited after 60 minutes.
- Each user may remove their own review.
- Each user may view all reviews for a listing.
- Each review must contain a timestamp.

Listing Images

- Each listing image may be edited by the seller.
- Each listing image must be uploaded by the seller.
- Each listing image may be removed by the seller.
- Each listing must contain at least one image but may contain multiple images.

Categories

- Each category must be created by an administrator.
- Each user may sort listings using categories.
- Each user may view listings from different categories.
- Each administrator may edit a category.
- Each administrator may delete a category.

Messages

- Each message must belong to one sender.
- Each message must contain a timestamp.
- A message may not be edited after 5 minutes.
- Each message may be deleted by the sender.

Live Chat

- Each user may send messages in the live chat.

- Each user may receive notifications from the live chat.
- Each user may mute the live chat on their device.
- Each user may unmute the live chat on their device.
- Users may not view live chat history older than 24 hours.

Favorites

- Each user may favorite a listing.
- Each user may remove a listing from their favorites.
- Each user may view their favorites on a separate page.

Marketplace

- Each user must become a verified seller before listing products.
- Each user may view any listed product.
- Each user may message a seller regarding their listing.
- Each user may view listings within their geographical area (optional).

Listings

- Each seller must be verified before creating a listing.
- Each seller must attach a product image, price, and description to each listing.
- Each seller must assign a category to their listing.
- Each seller must specify the condition of the product in the listing.
- If a listing is sold or unavailable, the seller must update its availability status.
- Each seller may remove a listing at any time.

Notifications

- Each user may receive one or more notifications.
- Each user may delete notifications.
- Each user may mark notifications as read.

Login

- Each user must register before logging in.
- Each user must log in before using the application.
- If a user forgets their password, they may recover or change it.
- If a user forgets their username, they may recover it.

Merchandise Store

- Each product listing must be created by an administrator.

- Each product listing must be edited by an administrator.
- Each product listing must contain a product image, price, and description.
- Each product listing must be visible to all users on the platform.
- Each product listing may only be removed by an administrator.

Announcements

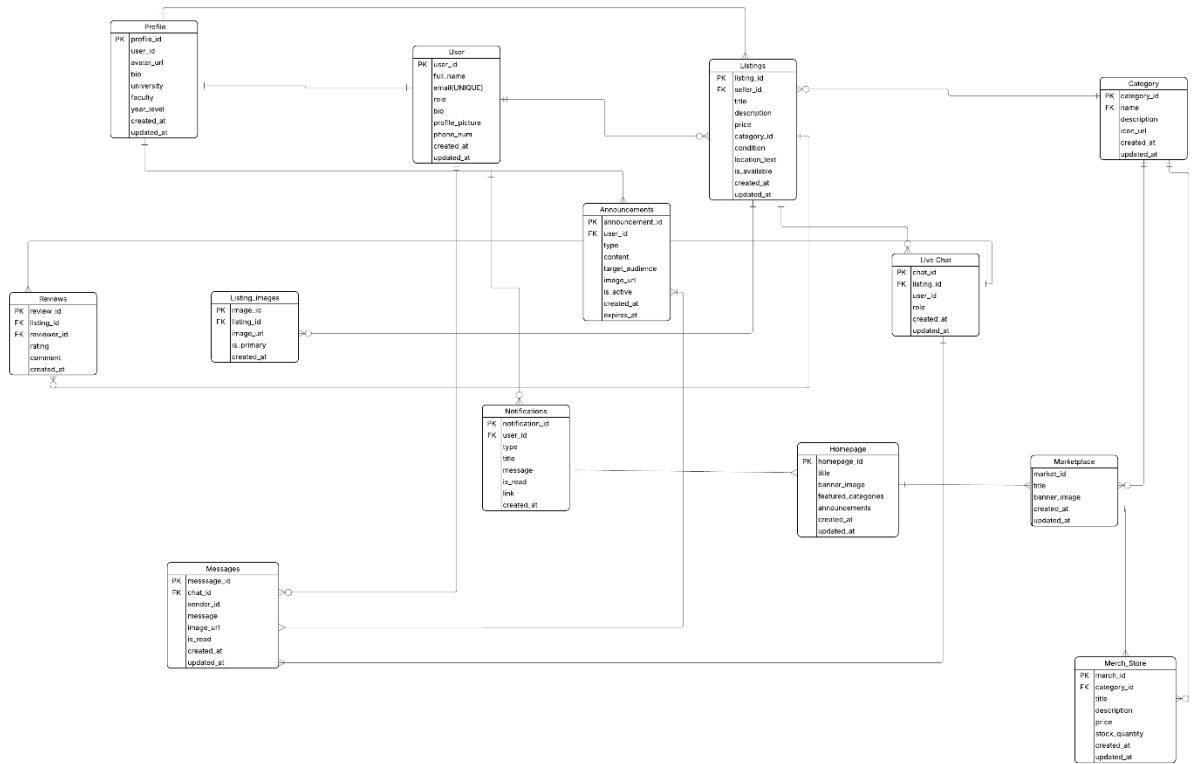
- Each announcement must be created by an administrator.
- Each announcement must be broadcast to all users simultaneously.
- Each user may view all announcements.
- Each announcement must contain a subject and a body.
- Each announcement may be scheduled for a future date and time.
- Each administrator may edit an announcement before it is published.
- Each administrator may delete an announcement.

Profile

- Each profile must belong to one user.
- Each profile must contain a user picture, name, surname, date of birth, geographical location, gender, student email, and student number.
- Each user may edit their own profile.
- Each user profile may be viewed by other users.
- Each user may delete their profile from the platform.
- A user may not edit another user's profile.

Home Page

- Each user may view the home page without logging in.
- Each user may view all active listings on the home page.
- Each user may view all announcements on the home page.
- Each user may search for product listings on the home page.
- Each user may filter product listings by geographical area on the home page (optional).
- Each user must log in before interacting with any listing on the home page.





Struktur: Dodekaeder silbernes

2nd Year

Information & Design

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innovative solutions and learning new things every day

University Cape Peninsula University of Technology

Faculty Informatics & Design

Year Level 3rd Year

Student ID: 219049678

Member Since 12 Aug 2022

Profile ID: PRF-0001245

Created At 12 Aug 2022, 10:30 AM

Updated At: 02 May 2024, 09:15 AM

2

Courses Enrolled 5

Assignments 12

Average Grade
78%

Unread Notifications
3



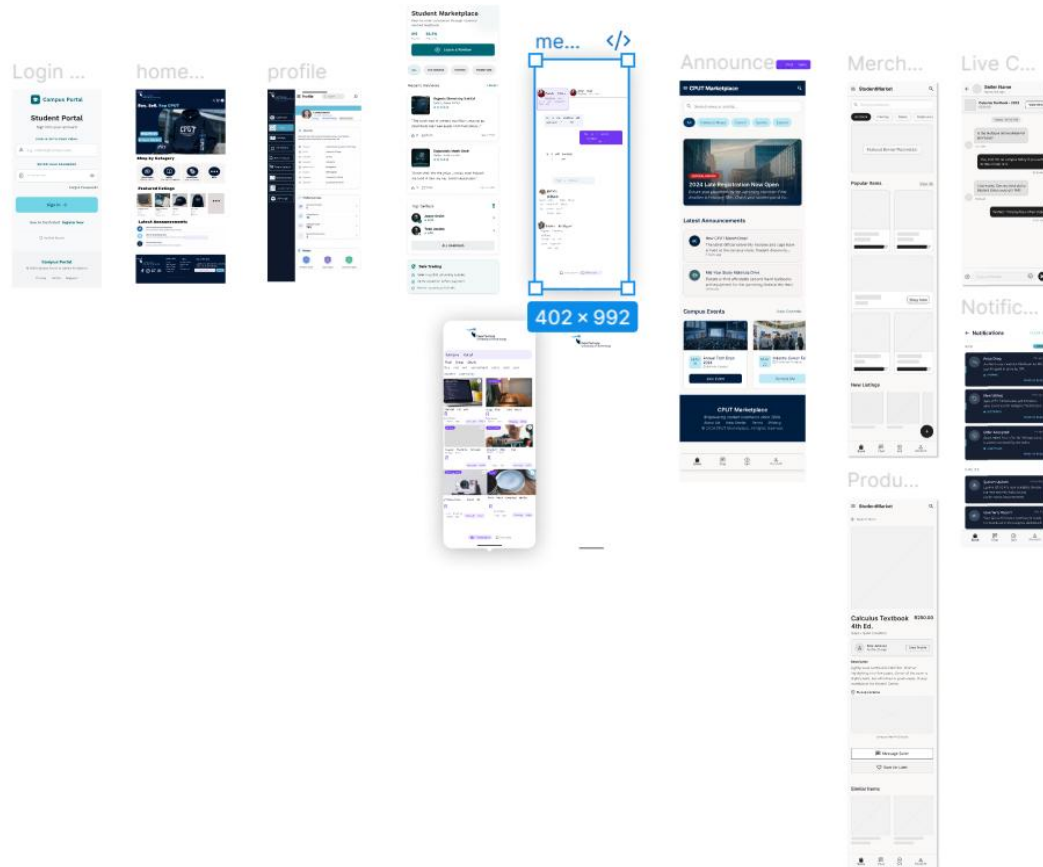
Problem Solver



Team Player



Tech Enthusiast



Reflection

For this phase of the project, my responsibility was the design and planning of the Profile Page for our Student marketplace system. My contribution included creating the wireframe in Figma, helping structure the ERD for the profile module, and analysing the business rules related to users, profiles, notifications, reviews, and messaging. The profile page was designed to give users a clean and user-friendly interface where they can manage and view their personal information.

When designing the wireframe in Figma, I focused on creating a layout that was modern, organised, and easy to navigate. The sidebar navigation was added to help users move quickly between sections such as Profile, Listings, Messages, Notifications, Marketplace, and Settings. I also included profile statistics, badges, and an “About Me” section to make the page feel more interactive and personalised. The design aimed to balance functionality and simplicity while still maintaining a professional appearance.

The ERD for the profile section was developed based on the system requirements and business rules provided by the group. I focused specifically on the entities related to the profile module, including User, Profile, Notifications, Messages, Reviews, and Listings. The relationships between these entities were important because the profile page interacts with multiple sections of the application. For example, one user can have many notifications, listings, and messages. I also ensured that the business rules supported the relationships shown in the ERD, such as each profile being linked to one user and profile information being editable. Shuttle tracking and

GPS-related entities were excluded from the project because those features would've made things too complicated to design as it would've been a whole sperate system.

One of the challenges I experienced was ensuring that the profile module connected properly with the rest of the application without overlapping too much with other members' work. I needed to make sure my section was detailed enough to show functionality, while still staying within the boundaries of the profile page responsibilities. Another challenge was designing the wireframe in a way that looked visually appealing but was also realistic and practical for implementation.

Through this process, I improved my skills in UI/UX design, database planning, and understanding entity relationships. I also gained more confidence using Figma to create structured wireframes and learned how business rules influence the design of an ERD. Overall, this task helped me understand how different parts of a system connect and the importance of planning before development begins. We have started created a few classes for the entities, but for now it's still a working progress.

Links

Wireframes: <https://www.figma.com/design/tULNjDqD4HvuGv11lao96q/The-groups-Wireframes-for-the-wireframes.?node-id=42-1334&t=VU5kVf44FI0ZFdn9-0>

Lucid for ERD: https://lucid.app/lucidchart/7344477d-ad94-40a5-a179-27c696ec4515/edit?invitationId=inv_d701e92d-461f-4a47-b7e3-9ff7489016f5&page=0_0#

GitHub:https://github.com/Jada-Rinquest/StaggeredSubs_T2_INV_222871296